



Adaptive Conversational Machine Learning Tutor using QReCC

Final Project Report

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1. Introduction

Conversational Artificial Intelligence systems have significantly improved in recent years due to advances in large language models, retrieval systems, and transformer architectures. However, most conversational question answering systems are primarily optimized for answer correctness rather than educational interaction quality.

In real educational environments, learning conversations involve much more than retrieving accurate answers. Effective tutoring systems must:

- understand conversational context
- interpret follow-up questions
- remember previous concepts
- detect student confusion
- adapt explanation style
- maintain coherent dialogue across multiple turns

Traditional QA systems often fail in these areas because they treat each query independently or rely on naive conversation concatenation.

The QReCC dataset provides an excellent benchmark for studying conversational question answering because it contains:

- conversational history
- context-dependent follow-up questions
- rewritten standalone queries
- retrieved evidence passages

This project proposes a lightweight adaptive conversational tutor specialized in machine learning education. Instead of focusing solely on answer correctness, the system models conversational educational interaction using:

- conversational memory
- semantic retrieval
- question rewriting
- confusion detection
- adaptive teaching strategies

The project demonstrates how conversational AI can support realistic educational dialogue while remaining lightweight and computationally efficient.

2. Problem Statement

Most conversational QA systems face several limitations:

2.1 Weak Conversational Memory

Many systems concatenate all previous conversation turns directly into the context window. This creates:

- noisy context
- irrelevant information
- scalability issues
- reduced retrieval precision

2.2 Poor Handling of Follow-up Questions

Educational conversations contain ambiguous queries such as:

- “Why?”
- “How does it work?”
- “Can you explain again?”

Without proper conversational understanding, retrieval systems fail to interpret these questions correctly.

2.3 Lack of Educational Adaptation

Most QA systems generate identical response styles regardless of:

- student understanding
- confusion level
- explanation preference

Educational tutoring requires adaptive behavior.

2.4 No Learning-State Modeling

Conventional conversational systems rarely model whether the student:

- understands concepts
- remains confused
- requires simplification
- prefers concise explanations

This limits educational effectiveness.

3. Project Objectives

The objectives of this project are:

1. Build a conversational machine learning tutor using QReCC-inspired conversational modeling.
2. Design a lightweight conversational system without relying on large expensive LLMs.
3. Improve dialogue coherence using intelligent conversational memory.
4. Improve question understanding using conversational question rewriting.
5. Detect student confusion signals.

6. Adapt teaching style dynamically.
7. Demonstrate educational interaction quality rather than only answer correctness.

4. Innovation Components

This project satisfies multiple innovation requirements from the assignment.

4.1 New Conversation Context Modeling

Instead of using complete dialogue history, the system dynamically retrieves semantically relevant memory.

This improves:

- conversational coherence
- retrieval quality
- context efficiency

4.2 Question Understanding Improvement

The tutor rewrites ambiguous conversational questions into standalone educational queries.

Example:

Original:

Why is it important?

Rewritten:

Why is gradient descent important in machine learning?

This improves semantic retrieval significantly.

4.3 Adaptive Teaching Behavior

The tutor dynamically changes teaching style depending on:

- confusion level
- explanation requests
- conversation context

Teaching modes include:

- beginner mode
- concise mode

- analogy mode
- detailed mode

4.4 Confusion Detection

The system detects confusion using linguistic patterns such as:

- “I don't understand”
- “Explain again”
- “Simplify”
- “What do you mean?”

The tutor then changes explanation strategy automatically.

4.5 Lightweight Conversational Intelligence

Instead of relying on large-scale LLMs, the project combines:

- semantic retrieval
- rule-based reasoning
- embedding-based memory
- adaptive response logic

This creates a lightweight but intelligent conversational tutor.

5. Dataset Analysis

5.1 QReCC Dataset

QReCC (Question Rewriting in Conversational Context) is a conversational question answering dataset designed for context-dependent query understanding.

The dataset contains:

- multi-turn conversations
- follow-up questions
- rewritten standalone queries
- retrieval passages

QReCC is particularly useful for:

- conversational retrieval
- context understanding
- question rewriting research
- dialogue coherence studies

5.2 Exploratory Data Analysis

Several observations were identified during EDA.

Observation 1 — Strong Context Dependence

Many questions relied heavily on previous turns.

Examples:

Why?

How?

What about that?

This motivated the conversational memory and rewriting modules.

Observation 2 — Ambiguous Follow-up Questions

A significant portion of conversational questions were ambiguous without dialogue history.

This motivated:

- question rewriting
- semantic memory retrieval

Observation 3 — Educational Clarification Patterns

Several conversational patterns resembled educational confusion:

- “I still don't understand”
- “Can you explain simply?”
- “Explain again”

This motivated the confusion detection module.

Observation 4 — Long Conversations

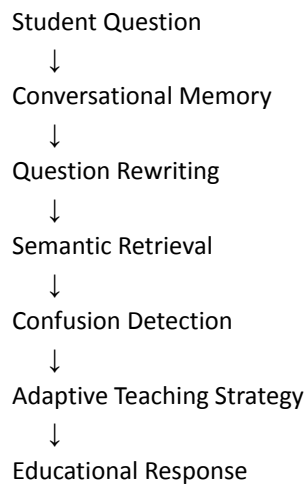
Long conversations introduced irrelevant context and retrieval noise.

This motivated:

- memory compression
- relevant context selection

6. System Architecture

6.1 Full Pipeline



7. System Components

7.1 Conversational Memory Module

Purpose

Maintain coherent multi-turn educational dialogue.

Problem

Using full conversation history introduces:

- noise
- irrelevant context
- retrieval degradation

Solution

The system retrieves semantically relevant memory using embedding similarity.

Implementation

- SentenceTransformer embeddings
- FAISS similarity search
- dynamic memory selection

Benefit

The tutor remembers only educationally relevant context.

7.2 Question Rewriting Module

Purpose

Transform ambiguous follow-up questions into standalone educational queries.

Example

Input:

Why is it important?

Output:

Why is overfitting important in machine learning?

Implementation

A hybrid approach was used:

- topic extraction
- rule-based rewriting
- conversational grounding

Motivation

Small generative models produced unstable rewrites during experimentation.

The hybrid approach improved:

- stability
- explainability
- reliability

7.3 Semantic Retrieval Module

Purpose

Retrieve educational knowledge relevant to the rewritten query.

Implementation

- SentenceTransformers embeddings
- FAISS vector similarity search
- educational knowledge base

Knowledge Base Topics

- machine learning
- overfitting
- gradient descent
- linear regression
- neural networks
- decision trees

Benefit

Focused educational retrieval improves consistency and coherence.

7.4 Confusion Detection Module

Purpose

Identify when the student does not understand the explanation.

Detection Signals

The system detects phrases such as:

- “I don't understand”
- “Explain again”
- “Simplify”
- “Still confused”

Implementation

- regex-based pattern matching
- lightweight rule-based detection

Benefit

Enables adaptive educational interaction.

7.5 Adaptive Teaching Module

Purpose

Change explanation style dynamically.

Teaching Modes

Mode	Description
Beginner	simple explanation
Concise	short technical answer
Detailed	expanded educational explanation
Analogy	real-world analogy

Example

Student:

I still don't understand gradient descent.

Tutor:

Imagine walking downhill while blindfolded...

Benefit

Improves educational interaction quality.

8. Technologies Used

Component	Technology
Programming Language	Python
Conversational Retrieval	FAISS
Embeddings	SentenceTransformers
Frontend	Streamlit
NLP	Transformers
Visualization	Matplotlib
Dataset	QReCC

9. Experimental Results

9.1 Conversational Coherence

The memory retrieval module improved contextual consistency by retrieving only semantically relevant dialogue turns.

Compared to naive history concatenation, the proposed approach:

- reduced irrelevant context
- improved topic tracking
- improved conversational continuity

9.2 Question Rewriting

Rewriting ambiguous questions significantly improved retrieval quality.

Examples:

Original Query	Rewritten Query
Why is it important?	Why is overfitting important in machine learning?
How does it work?	How does gradient descent work in machine learning?

The hybrid rewriting strategy produced more stable outputs than lightweight generative rewriting.

9.3 Adaptive Teaching

The tutor successfully changed teaching behavior depending on confusion signals.

Examples:

Student Input	Teaching Mode
Explain briefly	Concise
I still don't understand	Analogy
Explain in detail	Detailed

This demonstrated adaptive educational interaction.

10. Demo System

A Streamlit-based conversational tutoring application was developed.

Features

- interactive ML tutor
- conversational memory
- semantic retrieval
- adaptive tutoring
- tutor reasoning visualization
- confusion-aware explanations

Tutor Reasoning Panel

The demo visualizes:

- rewritten question
- detected teaching mode
- confusion detection
- retrieved memory
- retrieved educational knowledge

This makes the conversational intelligence transparent and explainable.